

Lab 2: 1-D Motion

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Objectives

In this experiment, you will

- Describe motion with constant acceleration,
- Understand the effect that error propagation has on measurement results,
- Discuss the advantages and disadvantages of using mathematical models of physical phenomena.

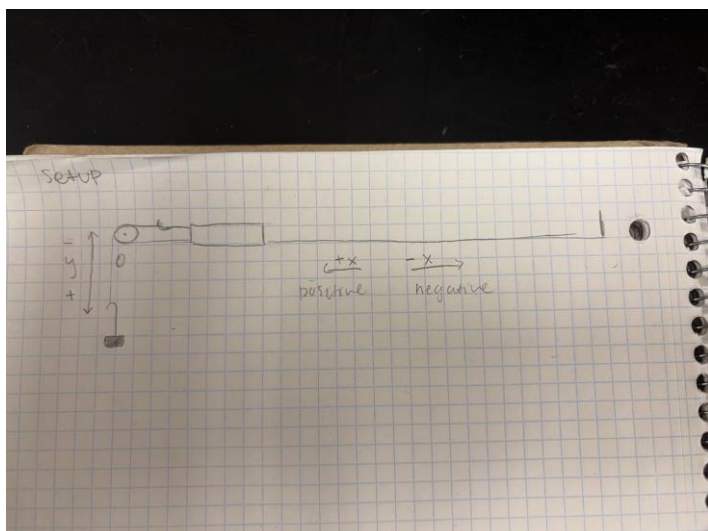
Apparatus

1. 1-2 m Track, Bumper
2. Wireless Smart Cart
3. Pulley with Clamp
4. Mass Hanger (~15 g)
5. String
6. Laptop
7. Bluetooth Adapter

A demonstration of the apparatus is set up on one of the lab tables for reference.

Setup

Set up the activity according to the directions in the “capstone instructions” (pdf).



Warm Up

For this part, do not attach any mass to the cart!

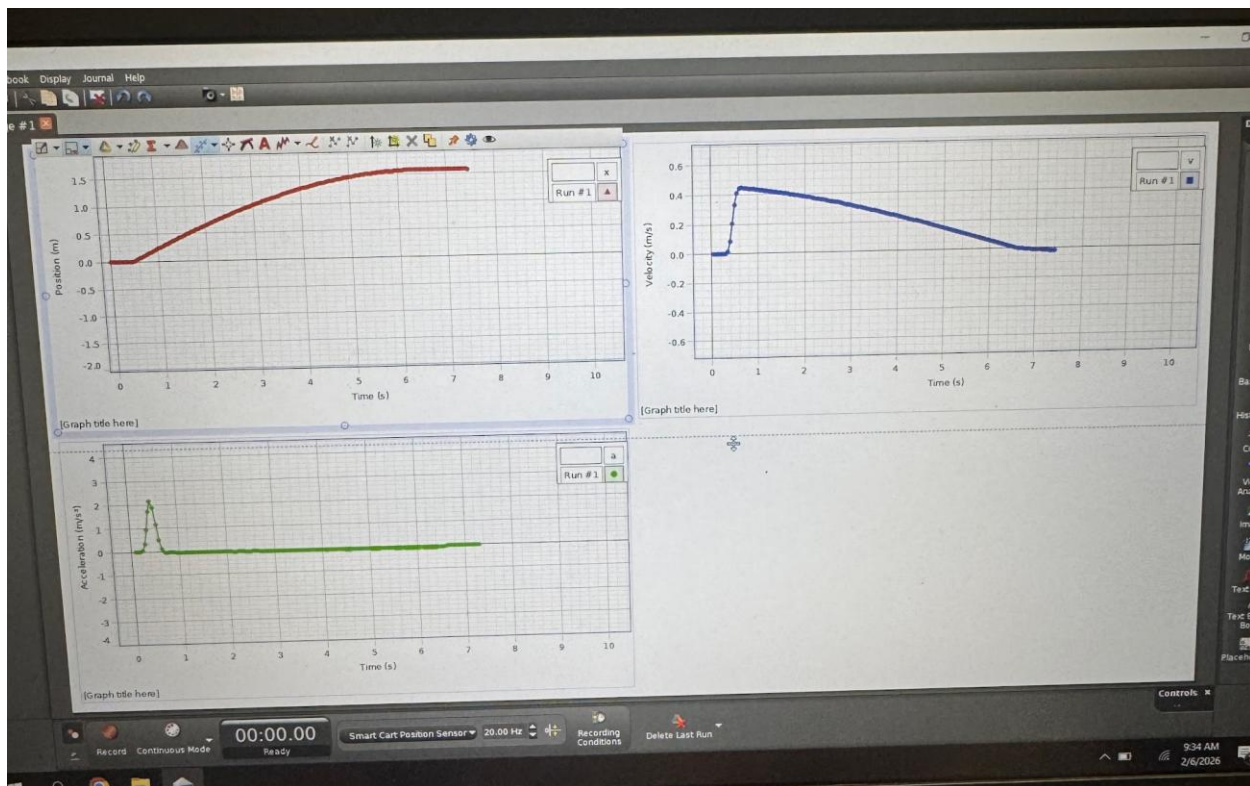
1. Open the position vs. time graph.
2. **Where is the “origin” of the sensor’s coordinate system?** Is it at one end of the track or somewhere in the middle? You and your group should decide where you would like to set the coordinate system’s origin to be used for your measurements. Depending on the equipment, you can “zero” the sensor to set the origin to be where you want.

We chose the end of the track closest to the pulley system at 0 cm as our origin for this experiment.

3. Give the cart a **gentle** nudge. *Start recording data **after** giving the cart a soft push and stop recording data **before** the cart reaches the end of the track.*

Based on the sensor’s data, what is the positive direction of the x-axis relative to the track and the sensor? Does this data confirm the location of the origin of the coordinate system?

The positive direction of the x-axis relative to the track and the sensor is back towards the pulley system. The negative direction of the x-axis is away from the pulley system, down the track, and down the tape measure.



Let the Phun begin

You are now ready to study the motion of the cart under the influence of a constant net force. A constant net force acting on an object will cause that object to move at constant acceleration. This means that the object will change its speed and/or direction of travel at a constant rate. In 1-D motion, this will cause the object to either speed up, slow down, or change direction along a straight line.

To study 1-D motion, set up the cart such that it is under the influence of a constant net force. You can do this in a number of different ways. In this lab, attach the cart to a mass and a pulley, as directed by the instructor.

- 1) Record Data. Starting from one end of the track, give the cart a gentle push such that it begins moving opposite to the direction of the constant net force acting on it. Eventually, the cart should reverse direction and return to its starting position. *Record data as soon as the cart starts to move on its own along the track and stop recording before it goes off the end of the track.*

2) Study the position vs. time graph.

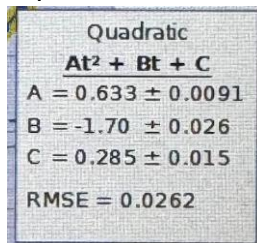
a. **What type of curve do you think best describes the cart's position vs. time graph (linear, quadratic, exponential, etc.)?**

- **quadratic**

b. Choose an interval in time to perform a "quadratic fit" approximation to the curve in the position vs. time graph (choose an interval that seems to be "least noisy").

What is the equation for the quadratic fit that approximates the position curve? Report the uncertainties as well.

Equation : $At^2 + Bt + C$



$$Y = (0.633 \pm 0.0091)(t^2) - (1.70 \pm 0.026)(t) + (0.285 \pm 0.015)$$

c. From the plot, what is the position of the cart at time $t = 0$? What is the value of C from the quadratic fit? How does C in this model differ from the actual starting position of the cart? From these data, is it reasonable to say that C represents the initial position of the cart? Why or why not?

Position of the cart at time = 0 is = to 0.

The $C = 0.285 \pm 0.015$ in the quadratic fit.

It is reasonable to say that the "C" value represents the initial position of the cart because we can assume that a value within 0.3 cm is reasonable due to human/experimental error.

d. If the values of C and the cart's actual starting position differ significantly, can we say that the mathematical model is "wrong"? Why or why not?

The mathematical model is not wrong, but it is the closest estimate or "best fit" for the data. The data includes human forces that can alter the constant motion of the cart.

e. To explore the answers to the previous questions, **pick some value for time and plug it into the quadratic fit equation.** Be sure to choose a time value that

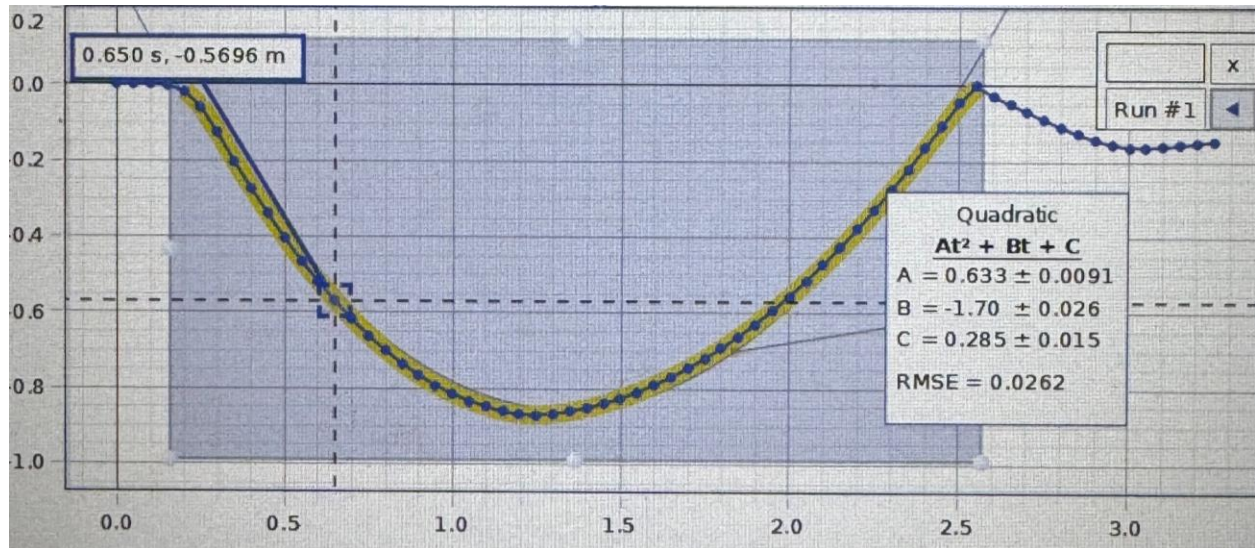
falls somewhere in the time interval during which the cart's motion. The result of this calculation will be the mathematical model's prediction for the cart's position at that time.

We chose the point (0.650s, -0.5696m) as our point. When we plugged this point into our quadratic equation, we got an answer of -0.5526 ± 0.0314

- f. Do a percent difference to compare the equation's predicted result with the measured position of the cart at that time. How well does the mathematical model for the cart's position match with the measurement of the cart's actual position? Is it reasonable to use this model to make predictions of the cart's position?

The percent difference of our measurements to the chosen point is a 3.0%





3) position vs. time and slope

- a. At which point or points is the slope of the tangent line at a point on the curve nearly horizontal? What was the cart doing during at that time (was it speeding up, slowing down, changing direction, stopped, etc.)?

The slope of the tangent line is nearly horizontal at near the lowest point of the position time graph, which is around 1.2-1.4 seconds. The cart's velocity was approximately zero, meaning the cart was momentarily stopped and changing direction.

- b. During which time interval(s) is the slope of the tangent line at any point on the curve negative? What was the cart doing during that time (was it speeding up, slowing down, changing direction, stopped, etc.)? In what direction was the cart traveling during this time?

The slope of the tangent line is negative from 0 to 1.2 seconds. During this time, the cart was moving in the negative direction (slowing down). Its velocity was negative but increasing toward zero until the cart momentarily stopped.

4) Study the velocity vs. time graph.

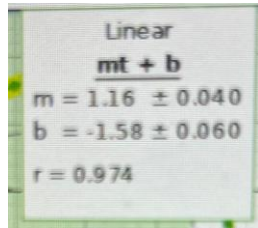
- a. What type of curve do you think best describes the cart's velocity vs. time graph (linear, quadratic, exponential, etc.)?

Linear

- b. Choose an interval in time to perform a “linear fit” approximation to the curve in the velocity vs. time graph (choose an interval that seems to be “least noisy”). What is the equation for the “best fit straight line” (linear fit) that approximates the velocity curve?

The equation is $mt + b$

$$y = (-1.16 \pm 0.040) + (-1.58 \pm 0.060)$$



- c. Based on your plot, what is the velocity of the cart at time $t = 0$? What is the value of b from the linear fit? How similar are these two values? What is their % difference? From these data, is it reasonable to say that b represents the initial velocity of the cart? Explain.

The velocity at $t = 0$ is equal to 0 m/s

$$b = -1.58 \pm .06$$

$$m = 1.16 \pm .04$$

The values of b and velocity at $t = 0$ are not similar; they are off by 1.58. The percent difference is 200%. $(1.58/(1.58/2)) * 100\% = 200\%$.

The values differ so greatly they differ by 200% therefore, it is not reasonable to say that b represents the initial velocity,

- d. When does the velocity curve cross the time axis? Let's call this value t_{cross} . What was the cart doing at t_{cross} ? What was the cart's speed at this time? How does the speed and time at this point compare to what you discovered in part a) of the position vs. time and slope question?

The velocity curve crosses the time axis at APPROX. 1.250s

This is at the point (1.250s, 0.011 m/s)

- e. During which time interval(s) is the velocity negative? What was the cart doing during this interval? How does this interval compare with the interval you found in part b) of the previous question for position vs. time and slope?

Our graph is flipped, so I will answer the question for the positive velocity. This was when the cart was being pulled back by the constant weight and was therefore moving backwards.

- f. Is the slope of the tangent line at a point on the velocity curve positive, negative, or nearly zero?

The slope is positive

- g. In which direction was the cart moving immediately before t_{cross} ? In which direction was the cart moving immediately after t_{cross} ?

Immediately before t_{cross} , the cart was moving along the y-axis in a negative direction. Immediately after t_{cross} , the cart was moving in the positive direction along the y-axis

5) Study the acceleration vs. time graph.

- a. Choose an interval of time in which to calculate the mean (average) acceleration. Is the mean acceleration positive, negative, or zero during this interval? How does the value for the mean acceleration compare with the value for the slope of the best fit straight line (from the linear approximation) that you obtained from part b) of the previous section? Compare by calculating the percent difference between these values. The time that I chose is between 0.5 and 2 seconds because the mean acceleration is positive and approximately 1.27 m/s^2 . This value closely matches the slope of the best-fit straight line from the velocity–time graph.

$$t = 0.5 \text{ s } v = -1.4 \text{ m/s}$$

$$t = 2.0 \text{ s } v = +0.5 \text{ m/s}$$

$$A_{\text{avg}} = 0.5 - (-1.4) / 2.0 - 0.5 = 1.9 / 1.5 = 1.27 \text{ m/s}^2$$

6) Comparing graphs

- a. Carefully compare the “noisiness” of each of the previous graphs (position vs. time, velocity vs. time, and acceleration vs. time). Note: I define “noisiness” to be the amount that a set of points fluctuates from the best fitting curve that models the data. Which graph appears “noisiest”? The noisiest graph is the acceleration graph, the noise comes from the variance of data points and additionally the spike due to the pull on the wheel and the string.
- b. Qualitatively rank the “noisiness” of each graph, from least noisy to most noisy. The least to most noisy graph is position vs time, velocity vs time, then acceleration vs time.
- c. What do you think the source of the noisiness is in each of the graphs and how do you think error propagation influences it? Hint: Even though the sensor measures the instantaneous position of the cart, it can only approximate instantaneous velocity and instantaneous acceleration by measuring the average velocity and the average acceleration of the cart over a short time interval. In light of this, think about how the uncertainty in position influences average velocity and how the uncertainty in average velocity influences average acceleration.

The position vs time graph is the least noisy because it has the lowest percent uncertainty. Additionally this is instantaneous which makes it more accurate. Instantaneous velocity or acceleration is not being measured only the average. This means that there is uncertainty in the positions for the Velocity and acceleration graphs. The sensor is directly measuring position vs time so the Velocity graphs linear fit is calculated by Position vs time graph which creates more error.

- d. What connection(s) do you see between the position vs. time and velocity vs. time graphs? What connection(s) do you see between the velocity vs. time and acceleration vs. time graphs?

The slope of the position vs time graph represents velocity, which explains the (quadratic) curved position graph, which is linear to the velocity time graph. And the velocity graph has a nearly constant positive slope, the acceleration vs. time graph is approximately constant and positive.

- e. How does the value B from the quadratic fit in the position vs. time graph compare with the velocity intercept b from the linear fit in the velocity vs. time graph? Is it reasonable to say that $B = v_0x$?

The value B from the position vs. time graph is -1.70 ± 0.026 , which is close to the B value of the velocity intercept of -1.58 ± 0.060 .

It is reasonable to say that $B = v_0x$ because, since velocity is the derivative of the position, the term must match the intercept in the equations.

- f. How does the value $2A$ from the quadratic fit in the position vs. time graph compare with the slope m from the linear fit in the velocity vs. time graph? How does $2A$ compare with the mean acceleration in the acceleration vs. time graph? Is it reasonable to say that $A = \frac{1}{2}a$?

The derivative of the quadratic function $= 2At + B$ which fits the format of the velocity vs time formula $(mt + b)$. Therefore, the $2A = m$, so they represent the same constant acceleration.

In the same way, $a = 2A$, so in the acceleration vs. time graph, the acceleration should equal $2A$ within the uncertainty.

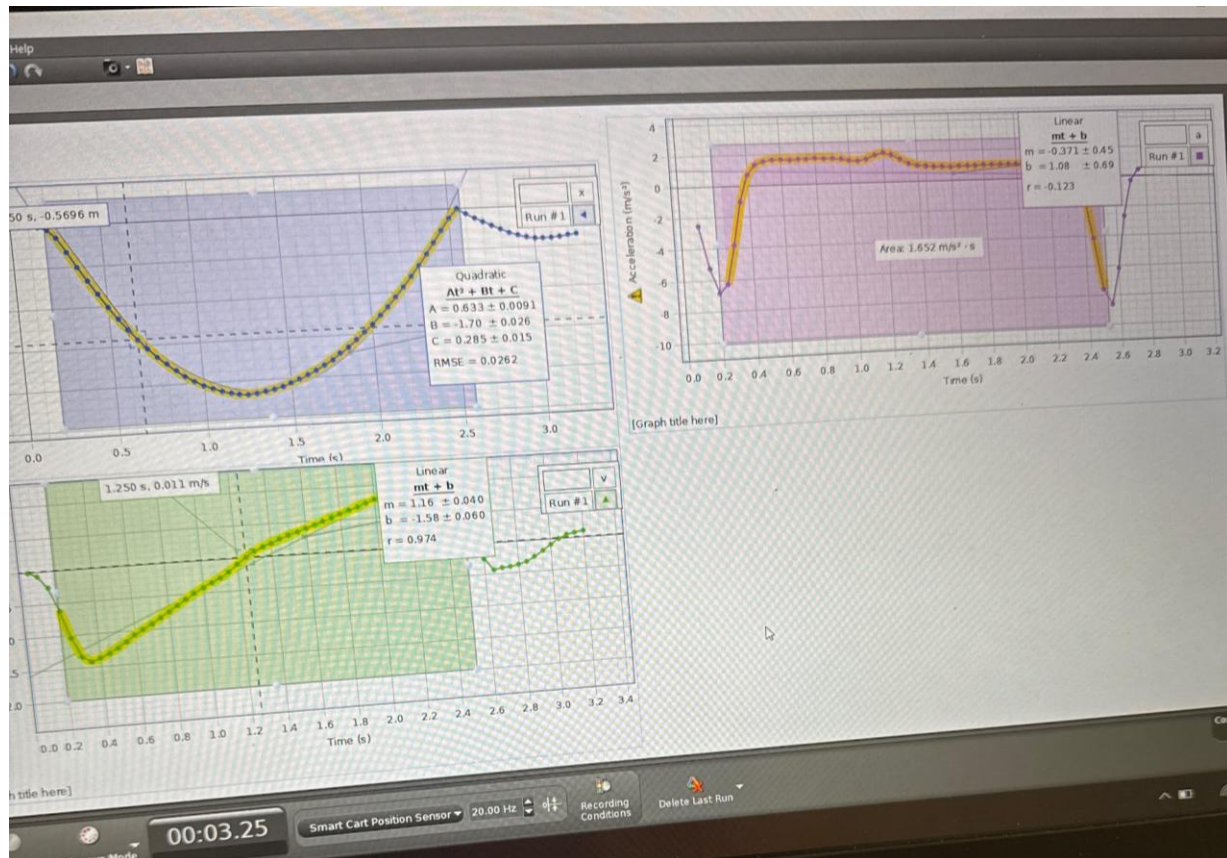
Yes, since $a = 2A$, then $\frac{1}{2}a = A$ due to division.

- g. Based on your two previous answers, rewrite the quadratic equation obtained from the position vs. time graph in terms of the quantities x , x_0 , v_0x , ax , and t , and rewrite the linear fit equation obtained from the velocity vs. time graph in terms of v_x , v_0x , ax , and t only.

$$A = \frac{1}{2} ax, B = v_0x, C = x_0, m = ax, b = v_0x$$

Position vs. Time → Quadratic equation = $x = x_0 + v_0x(t) + \frac{1}{2}ax(t^2)$

Velocity vs. Time → Linear Equation = $v_x = ax + v_0x(t)$



Putting it All Together

1. How did error propagation influence the reliability of the velocity vs. time and acceleration vs. time graphs? Each of the graphs have a different % error, this error slowly adds up and can hinder the reliability of the graphs. The graphs can be off by a couple of points due to this.
2. If you could pick only one graph from which to get all your data (position vs. time, velocity vs. time, or acceleration vs. time), which graph would you choose and why? To get all the data, the position v. time graph would be ideal since it shows in detail what the cart was doing and allows us to calculate everything such as acceleration and velocity.

3. If you look carefully at the position vs. time graph, you may notice that the “parabola” is not quite symmetric and may be skewed in one direction. If you look at the velocity vs. time graph, you may notice that the velocity curve is not quite linear, but has a “kink” in it such that the slope of the velocity line may change at the location of the kink. If you look at the acceleration vs. time graph, you may notice that the acceleration distinctly changes from one value to another value. If this is the case with your data, why do you think your graphs look the way they do? Is there something that can account for the way the graphs look that wasn’t considered in the mathematical models for the position, velocity, and acceleration?

Yes, the weight that is attached to the cart only starts pulling past a certain point due to some of the slack that is present in the string that attached the weight and the cart. The string initially has some slack but when it is pushed the string tightens and causes the cart to be slowed down by the weight.

4. What are some advantages, disadvantages, and limitations of using mathematical models to describe physical phenomena?

Some limitations of mathematical models is that they only take into account the data, they do not take into account outside forces such as air resistance or differences in how something is done. In a way mathematical models always assume that something for example a force acting upon the cart is constant which is not always true. Some advantages of mathematical models is they are very helpful in measuring, gathering data, and predicting a possible outcome. Some disadvantages are that the models are limited by data if the data is poor then the model will have trouble providing a correct answer